

# BÀI TẬP THỰC HÀNH

## PHÁT TRIỂN ỨNG DỤNG THIẾT BỊ DI ĐỘNG

### 12. Local Notification

**Bài tập 1.** Thiết kế ứng dụng sử dụng Notification cho phép người dùng subscribe và gửi thông báo về cho người sử dụng khi có tin mới

Yêu cầu:

- Khi người sử dụng chọn nút nhấn subscribe thì hệ thống phải lưu lại pushtoken vào csdl như firebase
- Người quản trị hệ thống sẽ gửi thông báo đến danh sách các user (pushtoken) khi có yêu cầu.

Bước 1. Tạo project MyLocalNotification

- `npx create-expo-app MyLocalNotification`

Bước 2. Cài đặt thư viện

- `npm install expo-notifications`
- `npm install expo-device`
- `npm install expo-constants`

Bước 3. Tạo Components\NotificationDemo.js

```
import React, { useState, useRef, useEffect } from 'react';
import { Platform, Text, View, StyleSheet, TextInput, Button, Image, Dimensions,
TouchableOpacity, ScrollView, Alert } from 'react-native';
import Ionicons from '@expo/vector-icons/Ionicons'

const NotificationDemo = ({ params, }) =>
{
  let [likes, setLikes] = useState(1000);
  const [token, setToken] = useState('not yet');
  const [notification, setNotification] = useState();

  useEffect(() =>
  {
    //step 1: getting a user's Expo Push Token

    //step2: register notification received.

  }, [likes]);

  const handle = async () =>
  {
    //step 3: call push notification
```

```

}

const screenHeight = Dimensions.get('window').height;
const screenWidth = Dimensions.get('window').width;

return (

  <View style={styles.container}>

    <Text style={styles.title}>Your pictures</Text>
    <Text>Token: {token}</Text>
    <View style={styles.box}>

      <Image style={styles.img}
        source={{ uri: 'https://images.hdqwalls.com/download/apple-mountains-lu-1280x1024.jpg' }} />

      <Image style={styles.img}
        source={{ uri: 'https://images.hdqwalls.com/download/lake-clouds-mountains-forest-landscape-scenic-reflection-jg-1280x1024.jpg' }} />

    </View>

    <TouchableOpacity onPress={() => { handle() }}>
      <View style={{ flexDirection: 'row', width: '100%' }}>
        <View style={styles.iconBoxBlue} >
          <Ionicons
            name='thumbs-up' size={24} color={'#FFF'} />
        </View>
        <View style={styles.iconBoxRed} >
          <Ionicons
            name='heart' size={24} color={'#FFF'} />
        </View>
        <Text style={styles.numLikes}> {likes}</Text>
      </View>
    </TouchableOpacity>

  </View>

);
}

const styles = StyleSheet.create({

```

```
container: {
  flex: 1,
  padding: 10,
  marginTop: 50,
},
title: {
  color: '#0096c7',
  justifyContent: 'flex-start',
  paddingVertical: 10,
  fontSize: 26,
  textAlign: 'left'
},
box: {
  flexDirection: 'row',
  justifyContent: 'space-between'
},
img: {
  width: 150,
  height: 300,
  margin: 10,
},
btn: {
  borderRadius: 5,
  alignItems: 'center',
  justifyContent: 'center',
  padding: 10,
  backgroundColor: 'blue',
  width: 200,
  color: '#FFF',
  marginTop: 10,
},
btnText: {
  color: '#FFF'
},
iconBoxRed: {
  width: 36,
  height: 36,
  borderRadius: 36 / 2,
  backgroundColor: '#d00000',
  alignItems: 'center',
  justifyContent: 'center',
},
iconBoxBlue: {
  width: 36,
  height: 36,
  borderRadius: 36 / 2,
```

```

        backgroundColor: '#0077b6',
        alignItems: 'center',
        justifyContent: 'center',
        marginHorizontal: 5
    },
    numLikes: {
        color: '#222',
        fontSize: 24,
    }
  })
export default NotificationDemo;

```

#### Bước 4. Thêm thư viện

```

import * as Notifications from 'expo-notifications';
import * as Device from 'expo-device';
import Constants from 'expo-constants';

Notifications.setNotificationHandler({
  handleNotification: async () => ({
    shouldShowAlert: true,
    shouldPlaySound: false,
    shouldSetBadge: false,
  }),
});

```

#### Bước 5. Xây dựng hàm getToken()

```

//step 1: getting a user's Expo Push Token
const getToken = async () =>
{
  let status;
  let result;

  status = await Notifications.getPermissionsAsync();
  if (status !== "granted")
  {
    result = await Notifications.requestPermissionsAsync();
    console.log(result);
  }
  if (result.status !== "granted")
  {
    alert("Failed to get push token for push notification!");
    return;
  } else
  {
    const projectId = Constants?.expoConfig?.extra?.eas?.projectId
    console.log(projectId)
    if (!projectId)

```

```

    {
      throw new Error('Project ID not found!!!');
    }
    let expoPushtoken = (await
Notifications.getExpoPushTokenAsync(projectId)).data;
    setToken(expoPushtoken);
    console.log('expoPushtoken',expoPushtoken);

  }
  if (Platform.OS === 'android')
  {
    Notifications.setNotificationChannelAsync('default', {
      name: 'default',
      importance: Notifications.AndroidImportance.MAX,
      vibrationPattern: [0, 250, 250, 250],
      lightColor: '#FF231F7C',

    });
  }
  return token;
};

```

#### Bước 6. Đăng ký notification

```

//step2: register notification received.
const subscription =
Notifications.addNotificationReceivedListener(notification =>
{
  setNotification(notification);
  console.log("notification received", notification)
  Alert.alert("Receive message",notification.request.content.body)
});

return () =>
{
  notificationListener.current?.remove();
  responseListener.current?.remove();

};

```

#### Bước 7. Gửi notification

```

const sendPushNotification = async () =>
{
  console.log("send notification ...", token)
  const message = {
    to: [token, token],
    title: 'Hey',
    body: 'Someone like your picture',
    data: { numberOfLikes: likes },
  };

  await fetch('https://exp.host/--/api/v2/push/send', {

```

```

    method: 'POST',
    headers: {
      Accept: 'application/json',
      'Accept-encoding': 'gzip, deflate',
      'Content-Type': 'application/json',
    },
    body: JSON.stringify(message),
  });
}

```

Bước 8. Xử lý gọi hàm gửi thông báo

```

const handle = async () =>
{
  //step 3: call push notification
  await sendPushNotification().then(() =>
  {
    console.log('send success');
    setLikes(++likes)
  });
}

```

Bước 9. Tạo TK trên trang

- <https://expo.dev/>

Bước 10. Cài đặt EAS CLI

- **npm install -g eas-cli**
- **eas login** → đăng nhập Tk đã tạo bước 9

Bước 11. Kích hoạt eas project eas build

- **eas build**

```

PS C:\Users\X1 gen 9\Desktop\MyDesktop\HK2531\PTUD DI DONG\mylocalnotification> eas login
Log in to EAS with email or username (exit and run eas login --help to see other login options)
✓ Email or username ... thi206.tranthitruong@gmail.com
✓ Password ... *****
Logged in
PS C:\Users\X1 gen 9\Desktop\MyDesktop\HK2531\PTUD DI DONG\mylocalnotification> eas build:configure
EAS project not configured.
✓ Would you like to automatically create an EAS project for @thi206/MyLocalNotification? ... yes
✓ Created @thi206/MyLocalNotification: https://expo.dev/accounts/thi206/projects/MyLocalNotification on EAS
✓ Linked local project to EAS project 6a8737cc-ab0b-4ad3-bfa6-a77dce23f309
💡 The following process will configure your iOS and/or Android project to be compatible with EAS Build. These changes only appl
y to your local project files and you can safely revert them at any time.

? Which platforms would you like to configure for EAS Build? » - Use arrow-keys. Return to submit.
> All
  iOS
  Android

```

Trong project → hệ thống sẽ tạo file eas.json và trong file app.json sẽ đính kèm projectId như sau:

```

{
  "expo": {
    "name": "MyLocalNotification",
    "slug": "MyLocalNotification",
    "version": "1.0.0",
    "orientation": "portrait",
    "icon": "./assets/icon.png",
    "userInterfaceStyle": "light",
    "newArchEnabled": true,
  }
}

```

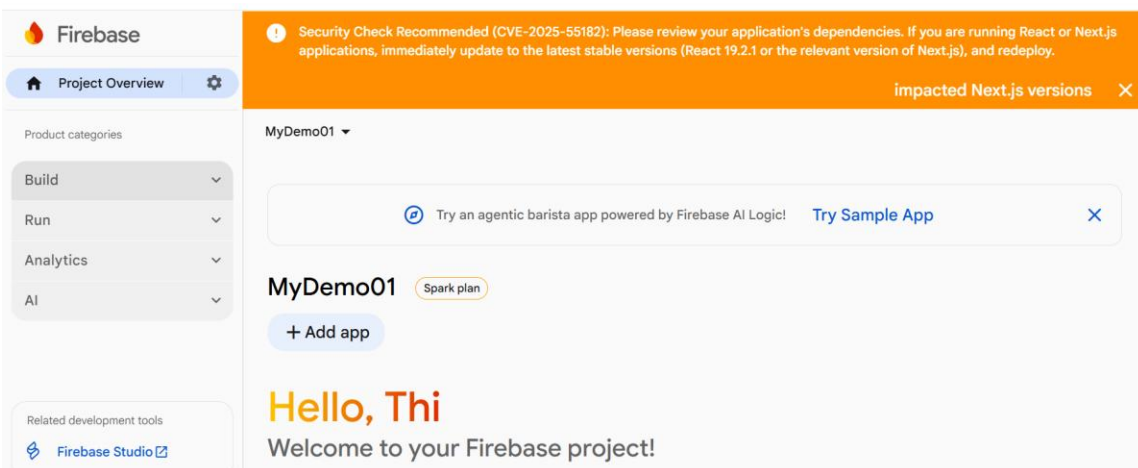
```

"splash": {
  "image": "./assets/splash-icon.png",
  "resizeMode": "contain",
  "backgroundColor": "#ffffff"
},
"ios": {
  "supportsTablet": true
},
"android": {
  "adaptiveIcon": {
    "foregroundImage": "./assets/adaptive-icon.png",
    "backgroundColor": "#ffffff"
  },
  "edgeToEdgeEnabled": true
},
"web": {
  "favicon": "./assets/favicon.png"
},
"extra": {
  "eas": {
    "projectId": "6a8737cc-ab0b-4ad3-bfa6-a77dce23f309"
  }
}
}
}
}

```

## Bước 12. Tạo project trên Firebase

- <https://console.firebase.google.com/?pli=1>
- Tạo project



- Add app android

## 1 Register app

Android package name ②

com.thi206.MyLocalNotification

App nickname (optional) ②

MyApp

## • Tải file google-services.json

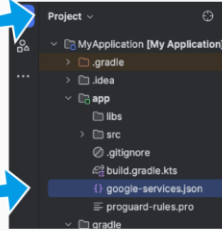
## 2 Download and then add config file

Instructions for Android Studio below | [Unity](#) [C++](#)

[Download google-services.json](#)

Switch to the Project view in Android Studio to see your project root directory.

Move your downloaded `google-services.json` file into your module (app-level) root directory.



Next

## Bước 13. Edit file app.json

```
{
  "expo": {
    "name": "MyLocalNotification",
    "slug": "MyLocalNotification",
    "version": "1.0.0",
    "orientation": "portrait",
    "icon": "./assets/icon.png",
    "userInterfaceStyle": "light",
    "newArchEnabled": true,
    "splash": {
      "image": "./assets/splash-icon.png",
      "resizeMode": "contain",
      "backgroundColor": "#ffffff"
    },
    "ios": {
      "supportsTablet": true
    },
    "android": {
      "adaptiveIcon": {
        "foregroundImage": "./assets/adaptive-icon.png",
        "backgroundColor": "#ffffff"
      },
      "edgeToEdgeEnabled": true,
      "package": "com.thi206.MyLocalNotification",
      "googleServicesFile": "./google-services.json"
    }
  }
}
```

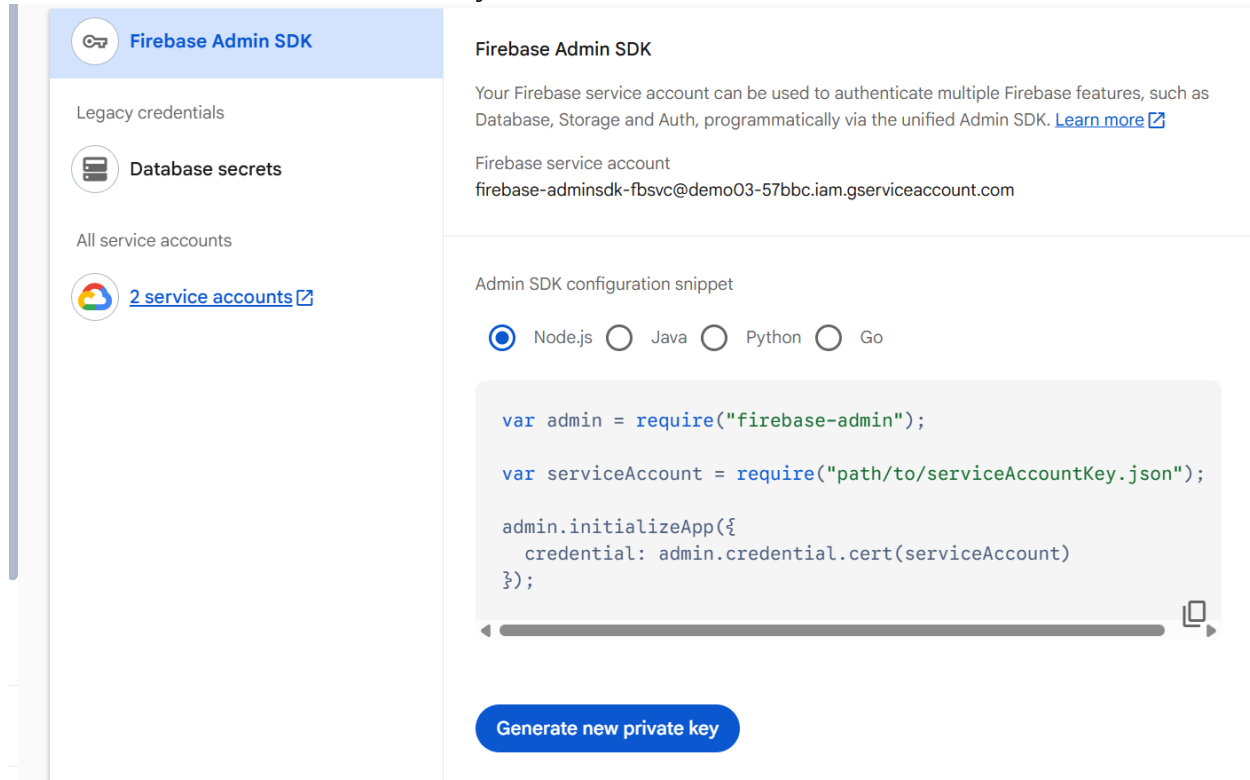


```

},
"web": {
  "favicon": "./assets/favicon.png"
},
"extra": {
  "eas": {
    "projectId": "6a8737cc-ab0b-4ad3-bfa6-a77dce23f309"
  }
}
}
}
}

```

## Bước 14. Tạo Service Account Key



**Firebase Admin SDK**

Your Firebase service account can be used to authenticate multiple Firebase features, such as Database, Storage and Auth, programmatically via the unified Admin SDK. [Learn more](#)

Firestore service account  
firebase-adminsdk-fbsvc@demo03-57bbc.iam.gserviceaccount.com

Admin SDK configuration snippet

☒ Node.js ☐ Java ☐ Python ☐ Go

```

var admin = require("firebase-admin");

var serviceAccount = require("path/to/serviceAccountKey.json");

admin.initializeApp({
  credential: admin.credential.cert(serviceAccount)
});

```

[Generate new private key](#)

## Bước 15. Tải file key lưu vào thư mục dự án

Chạy eas credentials

```
> TERMINAL
PS C:\2533\PTUDDIDONG\practices\mylocalnotification> eas credentials
None assigned yet

Push Notifications (FCM V1): Google Service Account Key For FCM V1
None assigned yet

Submissions: Google Service Account Key for Play Store Submissions
None assigned yet

Configuration: Build Credentials 3hSlkWrDPI (Default)
Keystore
Type                JKS
Key Alias            2d660d4298ecd493ba212c495510dd1b
MD5 Fingerprint      D1:FD:59:68:5C:3A:41:41:24:3E:9D:87:07:C9:3F:2E
SHA1 Fingerprint     8E:07:A5:28:20:17:29:EE:F2:B2:60:82:FE:8C:84:C1:7C:AE:BF:65
SHA256 Fingerprint   AD:2F:02:3D:35:80:4A:35:FF:A8:80:6C:B5:2F:59:29:E9:DE:0B:67:B3:E7:9C:21:DB:E1:72:6E:F6:5D:CC:7A
Updated              16 minutes ago

? What do you want to do? » - Use arrow-keys. Return to submit.
> Keystore: Manage everything needed to build your project
Google Service Account
Push Notifications (Legacy): Manage your FCM (Legacy) API Key
credentials.json: Upload/Download credentials between EAS servers and your local json
Go back
Exit
```

Bước 16. Chọn android

Bước 17. Chọn google service account

```
> TERMINAL
PS C:\2533\PTUDDIDONG\practices\mylocalnotification> eas credentials
Push Notifications (FCM Legacy)
None assigned yet

Push Notifications (FCM V1): Google Service Account Key For FCM V1
None assigned yet

Submissions: Google Service Account Key for Play Store Submissions
None assigned yet

Configuration: Build Credentials 3hSlkWrDPI (Default)
Keystore
Type                JKS
Key Alias            2d660d4298ecd493ba212c495510dd1b
MD5 Fingerprint      D1:FD:59:68:5C:3A:41:41:24:3E:9D:87:07:C9:3F:2E
SHA1 Fingerprint     8E:07:A5:28:20:17:29:EE:F2:B2:60:82:FE:8C:84:C1:7C:AE:BF:65
SHA256 Fingerprint   AD:2F:02:3D:35:80:4A:35:FF:A8:80:6C:B5:2F:59:29:E9:DE:0B:67:B3:E7:9C:21:DB:E1:72:6E:F6:5D:CC:7A
Updated              19 minutes ago

? What do you want to do? » - Use arrow-keys. Return to submit.
  Manage your Google Service Account Key for Play Store Submissions
  Manage your Google Service Account Key for Push Notifications (FCM V1)
> Upload a Google Service Account Key
Delete a Google Service Account Key
Go back
```

Upload file

Bước 18. Kiểm tra lại key đã được gán

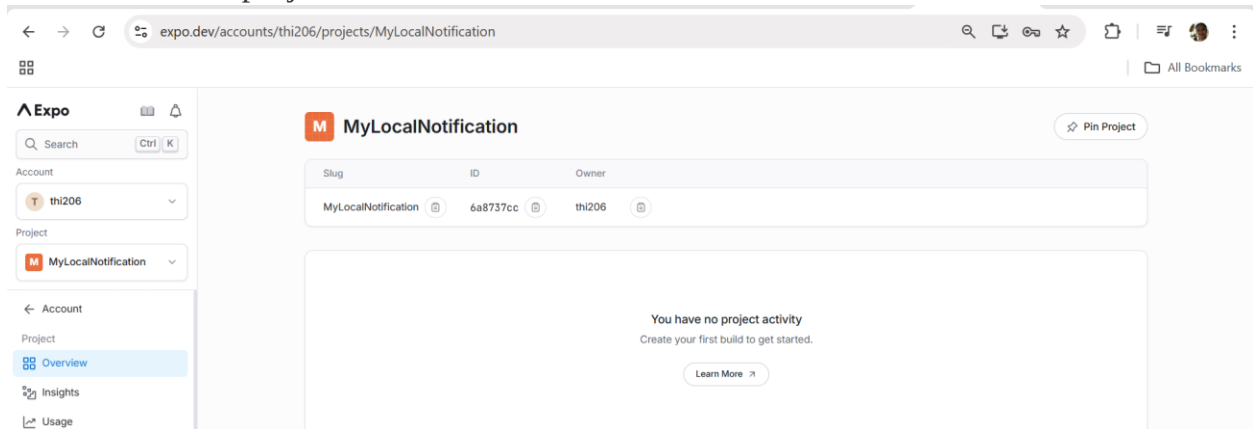
```
PS C:\2533\PTUDDIDONG\practices\mylocalnotification> eas credentials
Client Email    firebase-adminsdk-fbsvc@demo03-57bbc.iam.gserviceaccount.com
Client ID       101370679194658164931
Private Key ID  3c6ece130d6e09618eadfc9d45d24b1df1475936
Updated        5 minutes ago

Submissions: Google Service Account Key for Play Store Submissions
None assigned yet

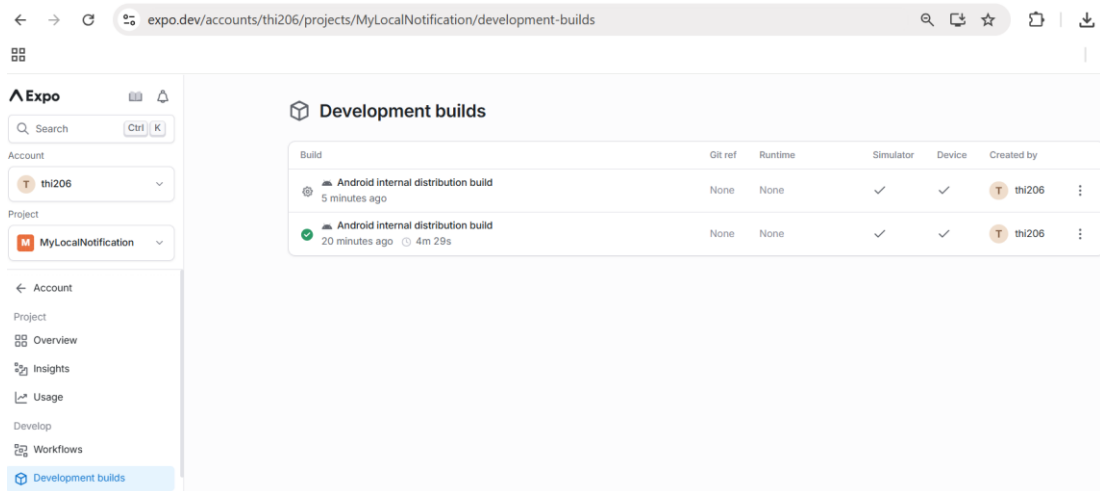
Configuration: Build Credentials 3hSlkwrDPI (Default)
Keystore
Type          JKS
Key Alias      2d660d4298ecd493ba212c495510dd1b
MD5 Fingerprint D1:FD:59:68:5C:3A:41:41:24:3E:9D:87:07:C9:3F:2E
SHA1 Fingerprint 8E:07:A5:28:20:17:29:EE:F2:B2:60:82:FE:8C:84:C1:7C:AE:BF:65
SHA256 Fingerprint AD:2F:02:3D:35:80:4A:35:FF:A8:80:6C:B5:2F:59:29:E9:DE:0B:67:B3:E7:9C:21:DB:E1:72:6E:F6:5D:CC:7A
Updated        45 minutes ago

? What do you want to do? » - Use arrow-keys. Return to submit.
Keystore: Manage everything needed to build your project
> Google Service Account
Push Notifications (Legacy): Manage your FCM (Legacy) API Key
credentials.json: Upload/Download credentials between EAS servers and your local json
Go back
Exit
```

## Bước 19. Chọn project → overview



- expo install expo-dev-client
- eas build --profile development --platform android → .apk hoặc .aab cho Android
- hoặc eas build --profile development --platform ios → ipa cho ios



## Bước 20. Chạy ứng dụng

- npx expo start --dev-client

**Bài tập 2.** Xây dựng ứng dụng My Reminder cho phép người dùng tạo các ghi nhớ đơn giản và cài đặt thời gian nhắc nhở sử dụng Notification

- Tạo nhắc nhở mới
- Tạo / xem danh sách các nhắc nhở
- Gửi nhắc nhở qua notification